

Sai Krishna Singapati

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Professional Summary

Embedded firmware engineer with 5+ years building real-time control systems, bootloaders, and networked embedded devices. I've taken products from board bring-up through production across industrial controls, mobile platforms, and automotive systems. Most comfortable working close to the hardware—whether that's debugging protocol issues with a logic analyzer or architecting firmware that'll run reliably for years. Currently focused on ARM Cortex-M platforms, industrial communication protocols, and setting up CI/CD pipelines that actually catch bugs before they ship.

Technical Skills

Languages Embedded C, C++, Python	Platforms ARM Cortex-M, STM32, ESP32, Cypress, DSPIC, Raspberry Pi	RTOS FreeRTOS, Zephyr, CMSIS-RTOS
Protocols TCP/IP, CAN, SPI, I2C, UART, RS485, BLE, Modbus, BACnet	Tools JTAG, Logic Analyzers, Oscilloscopes, Wireshark	DevOps Git, Jenkins CI/CD, CMake, GTest, Valgrind, GCOVR

Work Experience

Firmware Developer II | Q-PAC Systems, Elkton, FL

August 2023 – Present

- Built an Ethernet-based dual-bank bootloader for STM32H5, enabling secure over-the-network firmware updates with rollback capability.
- Integrated industrial communication stacks (IP and serial variants) for multi-protocol device interoperability.
- Developed firmware for a real-time motor control system managing an array of EC motors, focusing on fault tolerance and smooth operation under varying loads.
- Led board bring-up and DVT testing; designed a layered firmware architecture that made it straightforward to add new sensor types and communication interfaces.
- Set up a Jenkins CI/CD pipeline with automated builds, unit tests (GTest), and static analysis—caught several memory issues before they hit hardware.

System/Software Test Engineer | Qualcomm, San Diego, CA

July 2021 – July 2023

- Tested camera subsystem drivers on mobile chipsets using Google's CTS/VTs/ITS conformance suites.
- Wrote Python automation scripts that cut manual test time significantly and improved coverage consistency.
- Debugged Linux kernel-level camera driver issues, working closely with firmware teams to isolate timing and memory problems.
- Contributed to design reviews and pushed for process improvements that reduced regression cycles.

Firmware Engineer | DURA Automotive, Hyderabad, India

December 2018 – May 2019

- Performed unit testing and validation for automotive gear shifter firmware using LDRA, catching edge cases in state machine logic.
- Gained hands-on experience with automotive SDLC practices, requirements traceability, and version control workflows.

Education

M.S. Electrical & Computer Engineering (Embedded Systems)

San Diego State University • GPA: 3.67/4.0 • August 2019 – May 2021

Relevant: Embedded RTOS, Computer Networks, Cyber-Physical Systems, Machine Learning

B.E. Electronics & Communication Engineering

Vellore Institute of Technology • GPA: 7.86/10 • August 2014 – May 2018

Relevant: C Programming, Data Structures, Microprocessors & Microcontrollers, Digital System Design, Computer Architecture, Computer Networks